Forensic Tools & Documentation

Digital Forensics and Evidence Management Program

1. Project Overview

The project implements a digital forensic investigation workflow designed to identify, collect, preserve, examine, recover, and document digital evidence.

The objective is to establish a repeatable forensic examination process that maintains evidence integrity, supports chain-of-custody tracking, and produces professional investigation reports.

A completed forensic investigation, **Digital Evidence Analysis: Case 001**, demonstrates the practical implementation of multiple forensic tools and evidence-analysis techniques.

The investigation included:

- File-system examination.
- Deleted-file identification and recovery.
- Data carving using independent forensic tools.
- File metadata examination.
- Timeline reconstruction.
- Windows Registry analysis.
- Correlation of independent forensic artifacts.
- SHA-256 and MD5 integrity verification.
- Evidence registration and chain-of-custody documentation.

The investigation used a controlled Linux ext4 disk image and a separate NIST Windows Registry dataset.

2. Forensic Environment and Tools

The forensic environment was established using a macOS host running an ARM64 Parrot Security OS 7 virtual machine through UTM.

The examination environment included the following tools:

Tool	Version	Primary functionality	Status
Autopsy	2.24-6	Forensic case management, file examination, deleted-file analysis, keyword search, timelines	Completed

Sleuth Kit	4.12.1	File-system analysis, metadata examination, inode investigation, timeline generation	Completed
Foremost	1.5.7	Deleted-file recovery through signature-based carving	Completed
PhotoRec	7.2	Independent file recovery through signature analysis	Completed
sha256sum / md5sum	System utilities	Evidence integrity verification	Completed
reglookup	1.0.1	Windows Registry hive examination	Completed

The project successfully used four forensic tools, including Autopsy, which is explicitly identified in the rubric.

3. Forensic Case Management

A forensic case was created using Autopsy to organize and examine the digital evidence.

Case information

- Case name: `Digital_Evidence_Analysis_Case001`
- Report title: Digital Evidence Analysis Report: Case 001
- Investigator: Drew
- Examination date: November 3, 2026
- Environment: Parrot OS 7 ARM64

Autopsy provided a graphical forensic examination environment with functionality for:

- File and directory listing.
- Deleted-file identification.
- File metadata examination.
- Keyword searches.
- Unallocated-space examination.
- Timeline visualization.

The case incorporated a controlled 256 MB ext4 forensic disk image containing sample documents, a log file, and two deliberately deleted files.

Autopsy successfully imported the verified working image and independently displayed the file-system structure, deleted inodes, and timeline information.

The custom evidence-management system complements Autopsy by maintaining evidence identifiers, integrity hashes, verification records, and chain-of-custody information.

4. Evidence Acquisition and Preservation

A controlled forensic disk image was created to provide a known environment for testing evidence examination and recovery techniques.

The image was created using:

```
None
dd if=/dev/zero of=forensic_image.dd bs=1M count=256
```

The image was formatted using `mkfs.ext4` and mounted through loopback.

Sample files were created approximately five seconds apart to provide distinguishable timestamps. Two files were subsequently deleted to simulate a forensic recovery scenario.

After the deletion operations:

- The image was synchronized.
- The filesystem was unmounted.
- The original image was hashed.
- The original image was made read-only using `chmod 444`.
- A separate working copy was created.
- All subsequent analysis was performed on the working copy.

4.1 Integrity verification

The original image and working copy produced identical SHA-256 values.

Evidence	SHA-256
Original forensic_image.dd	`d7112134ca40c4003600d932ded3dd9a480029fb40cf1987b0d38e2eba265d6a`
Working forensic_working.dd	`d7112134ca40c4003600d932ded3dd9a480029fb40cf1987b0d38e2eba265d6a`

Both images also produced the same MD5 hash:

`889728cb91bec76dd0dd3ffd040633d3`

Autopsy independently calculated the same MD5 value when the image was imported.

This confirmed that Autopsy examined the same verified working copy.

5. File-System Examination Using Sleuth Kit

Sleuth Kit was used to investigate the ext4 filesystem and identify existing and deleted files.

The examination included:

Utility	Purpose
`fsstat`	Examine filesystem properties and metadata
`fls`	List files and directories
`istat`	Examine inode metadata
`icat`	Attempt recovery using inode metadata
`blkstat`	Examine block allocation status
`blkcat`	Extract content from filesystem blocks
`mactime`	Generate a chronological timeline

5.1 Filesystem examination

The `fsstat` examination identified:

- Filesystem type: ext4.
- Block size: 1,024 bytes.
- Total inodes: 65,537.
- Filesystem creation and first mount: 19:16:58 UTC.
- Last filesystem write: 19:23:26 UTC.

The `fls` utility identified three directories:

- Documents
- Logs
- Downloads

Three live files were also identified.

A deleted-file examination using `fls -r -d -p` identified two deleted inodes:

- Inode 32775.
- Inode 32776.

These were associated with the original files `deleted_evidence.txt` and `deleted_report.pdf`.

5.2 Metadata examination

The `istat` utility was used to examine the deleted inodes.

Both deleted files had zero-length inode metadata and no remaining block pointers.

The `icat` utility consequently returned empty files.

This demonstrated a limitation of metadata-based recovery when the filesystem has cleared the deleted file's extent pointers.

The investigation therefore proceeded using alternative recovery methods.

6. Deleted-File Recovery

The investigation used Foremost, PhotoRec, Sleuth Kit, and Autopsy to demonstrate different approaches to recovering deleted information.

6.1 Foremost recovery

Foremost was used to perform signature-based PDF carving.

The command was:

```
None
foremost -v -t pdf
```

The examination successfully recovered one PDF file.

Results:

- Recovered file size: 142 bytes.
- Recovery offset: 143,132,672 bytes.
- Filesystem block: 139778.
- Recovered file type: PDF 1.4.

The `blkstat` utility confirmed that the associated block was unallocated.

6.2 PhotoRec recovery

PhotoRec was used as an independent recovery tool.

The examination searched free space and recovered:

`f0279556.pdf`

Results:

- Recovered files: 1.
- File size: 142 bytes.
- Sector: 279556.
- Recovery location: 143,132,672 bytes.

The recovered file originated from the same location identified by Foremost.

The SHA-256 values of both recovered PDFs matched:

`54725981e65c4cb6635d6e2faa8ac15070c84f169b66d21aa669a8ceedca939c`

This demonstrated independent agreement between the two file-carving utilities.

6.3 Text-file recovery

A separate deleted text file was recovered using keyword searching and block extraction.

Because plain-text files do not necessarily contain recognizable file signatures, traditional signature-based carving was not suitable for this artifact.

The `strings` utility identified the following content:

```
Deleted evidence recovery test - Document 001
```

The content was located at byte offset 144,704,512, corresponding to block 141313.

The `blkstat` utility confirmed that the block was unallocated.

The `blkcat` utility was used to extract the 46-byte content.

Autopsy's keyword search of unallocated space independently identified the same content in fragment 141313.

The recovered text was consistent with the original deleted file based on its size, content, and storage location. The original filename and inode relationship could not be definitively re-established after deletion.

6.4 Recovery comparison

Recovery method	Target	Result
Sleuth Kit `icat`	Deleted text and PDF	Unsuccessful; returned zero-byte files
Foremost	Deleted PDF	Successfully recovered
PhotoRec	Deleted PDF	Successfully recovered
Sleuth Kit `strings` / `blkcat`	Deleted text	Successfully recovered
Autopsy keyword search	Deleted text	Independently identified

The investigation demonstrated that metadata-based recovery can fail when filesystem pointers are cleared, while content-based recovery can succeed when the underlying data remains in unallocated space.

7. Autopsy Examination and Validation

Autopsy was used to create the forensic case and independently examine the working image.

The examination included:

- Case creation.
- Forensic image import.
- File-system listing.
- Deleted-file identification.
- Inode examination.
- Timeline visualization.
- Keyword searching of unallocated space.

Autopsy displayed deleted inodes 32775 and 32776 under All Deleted Files.

It also identified their deletion timestamps:

- Inode 32775: 19:22:43 UTC.
- Inode 32776: 19:22:54 UTC.

Autopsy's timeline view produced the same overall event sequence as the Sleuth Kit timeline.

The keyword search independently located the deleted text content in fragment 141313.

This provided independent validation of findings produced through command-line examination.

8. Timeline Reconstruction

The project reconstructed a timeline of filesystem activity using Sleuth Kit.

The process used:

```
None
fls -r -m "/" -z UTC
```

The resulting bodyfile was processed using:

```
None
mactime -b <bodyfile> -d -y -z UTC
```

The bodyfile contained 10 lines and the resulting timeline contained 19 lines.

Both files were hashed for integrity verification.

8.1 Reconstructed event timeline

Time (UTC)	Event
19:16:58	Filesystem created and first mounted
19:17:41	Documents, Logs, and Downloads directories created
19:18:09	investigation.txt created
19:18:37	access.log created
19:19:04	account_changes.txt created
19:20:25	deleted_evidence.txt created
19:20:44	deleted_report.pdf created
19:22:43	Inode 32775 deleted
19:22:54	Inode 32776 deleted
19:23:05	Downloads directory accessed
19:23:26	Filesystem last written

19:24:38 Images hashed and original locked read-only

The timeline was also examined through Autopsy.

The agreement between Sleuth Kit and Autopsy supported the reconstructed sequence.

9. Windows Registry Forensics

A separate Windows Registry investigation was conducted using the NIST CFReDS Windows Registry dataset.

The dataset represented a Windows 10 environment, build 10586.

The investigation examined registry hives and correlated their contents with a scripted action log and process-monitor-style event logs.

9.1 Evidence preservation

The downloaded archive was:

`cfreds-2017-winreg_sgrd-10.7z`

Its SHA-256 hash was recorded:

`1d5ebaadb64ebf92b0234db7f27eb376dc818de263b186c8a2fda8640b88f610`

The archive extracted successfully into:

- 983 files.
- 181 folders.

All extracted files were copied into a working directory and hashed.

The resulting hash-list file was also preserved and the originals were made read-only.

9.2 Registry examination

The `reglookup` 1.0.1 utility was used to examine registry hives from the working copy.

The investigation correlated three scripted user actions against registry timestamps and process event logs.

Findings included:

Finding 1: Local user account creation

The investigation identified the creation of a CFTT user account.

The account was associated with RID `0x3EB` (1003).

The action log, process event log, and SAM registry hive provided matching evidence of the account creation process.

The registry and event-log timestamps agreed within approximately eight seconds.

Finding 2: Administrator group membership

The investigation identified an event associated with adding the CFTT account to the Administrators group.

The process event log and registry artifacts supported the relationship.

However, the registry utility did not decode the SAM binary group-membership values, so the conclusion was based on correlated event and registry evidence rather than a fully decoded membership record.

Finding 3: User Account Control configuration

The action log indicated that UAC was disabled.

However, the registry showed:

- `EnableLUA = 1`
- `ConsentPromptBehaviorAdmin = 0`
- `PromptOnSecureDesktop = 0`

These values were consistent with administrator prompts being disabled rather than UAC being completely disabled.

The finding was therefore documented as consistent with the scripted action, not as definitive proof that UAC was disabled.

9.3 Registry investigation limitations

The investigation identified several limitations:

- The dataset was scripted and included a known action log.
- Registry transaction log files were not replayed.
- SAM binary values were not fully decoded.
- Some additional accounts were not examined.
- Only three scripted actions were correlated.

The registry investigation demonstrates the ability to correlate system activity with persistent registry artifacts while documenting uncertainty and technical limitations.

10. Artifact Correlation and Findings

The investigation correlated information from multiple forensic sources.

The primary findings included:

Finding 1: Inode continuity

The inode numbers recorded before deletion matched the orphaned inodes identified during subsequent examination.

The recovered PDF also matched the original pre-deletion file size.

Finding 2: Independent tool agreement

Sleuth Kit and Autopsy produced matching timeline sequences.

Foremost and PhotoRec recovered bit-identical PDF files from the same storage location.

Finding 3: Metadata and log discrepancy

The access.log file contained a timestamp of 10:15:00 on November 3, 2026, while its filesystem creation time was 19:18:37 UTC.

The difference demonstrated why file contents and filesystem metadata should be examined together.

Finding 4: Recovered text correlation

The recovered text was consistent with the deleted_evidence.txt file based on content, length, and storage location.

However, the original filename and inode relationship could not be definitively established.

Finding 5: Windows account activity

The Windows Registry investigation correlated the creation of the CFTT account and associated administrative activity across the action log, process event log, and registry hives.

The findings demonstrate the value of correlating independent forensic artifacts rather than relying on a single data source.

11. Evidence Register and Chain of Custody

The project incorporates a structured evidence register to maintain traceability and document evidence integrity.

The Case 001 report includes the following evidence records:

Evidence ID	Item
EV-001	Original forensic disk image
EV-002	Verified forensic working copy
EV-003	Foremost recovered PDF
EV-004	PhotoRec recovered PDF
EV-005	Recovered text artifact
EV-006	Filesystem bodyfile
EV-007	Reconstructed timeline
EV-008	Autopsy unallocated-space extraction
EV-009	NIST Windows Registry dataset
EV-010	Registry dataset hash list
EV-011	Working copy of extracted registry dataset

The evidence records include hash information, timestamps, and investigator information.

The wider project evidence-management implementation also maintains:

- `evidence/logs/chain_of_custody.csv`
- `evidence/logs/verification_log.csv`

These records support evidence handling, verification, and audit activities.

Documentation item requiring completion: The Case 001 report identifies that exact recording times for some evidence entries must be filled in from terminal history before final submission.

12. Formal Forensic Reporting

A formal investigation report was produced:

Digital Evidence Analysis Report: Case 001

The report contains:

- Investigation overview.
- Tool and environment information.
- Evidence acquisition and preservation.
- File-system examination.
- Deleted-file recovery results.
- Timeline reconstruction.
- Artifact correlation.
- Windows Registry analysis.
- Evidence register.
- Chain-of-custody information.
- Technical limitations.
- Investigation conclusion.
- Screenshot and figure index.

The report includes 33 indexed screenshots covering tool installation, tool versions, evidence creation, image hashing, file-system examination, recovery operations, timeline reconstruction, Autopsy analysis, and Windows Registry examination.

The reporting methodology distinguishes observed evidence from interpretations and acknowledges when an artifact cannot be conclusively attributed to a particular source or action.

13. Validation and Testing

The project performed multiple validation activities to demonstrate the reliability of its forensic workflow.

Completed validation includes:

- SHA-256 comparison of original and working forensic images.
- MD5 comparison with Autopsy's calculated image hash.
- Independent timeline examination using Autopsy and Sleuth Kit.
- Independent PDF recovery using Foremost and PhotoRec.
- Hash comparison of recovered PDF files.
- Comparison of recovered text against known sample content.
- Verification of unallocated storage blocks.
- Correlation of Windows Registry timestamps with event logs.
- Documentation of technical limitations and uncertain findings.

The recovered PDF files were bit-identical, and their SHA-256 hashes matched a regenerated copy of the known original content.

The recovered text also matched the original sample string.

These tests were possible because the investigation used controlled evidence with known ground truth.

14. Limitations and Future Development

The investigation was conducted using controlled evidence rather than a live production workstation.

The primary limitations include:

- The Linux image was a simulated ext4 filesystem.
- Deleted-file recovery depended on the original data blocks remaining unallocated and not being overwritten.
- The original filenames and inode relationships for deleted files were not fully recoverable.
- The recovered text's source attribution was circumstantial.
- The Windows Registry investigation used a separate NIST reference dataset.
- The registry examination did not replay transaction logs or fully decode SAM binary values.
- Autopsy 2.24 presented a parsing warning for a synthetic orphan-file entry.
- Exact recording times remain to be completed for some evidence-register entries.

Additional development opportunities include:

- Expand forensic case management and reporting.
- Implement FTK Imager for acquisition and verification.
- Introduce Volatility for memory forensics.
- Evaluate SIFT Workstation as an additional forensic environment.
- Expand evidence auditing and automated reporting.
- Improve the integration between forensic case records and the custom evidence-management system.

15. Conclusion

The project successfully implemented and demonstrated multiple digital forensic tools and investigation techniques.

Four forensic tools were used in Case 001:

1. Autopsy for case creation, file examination, deleted-file identification, keyword searching, and timeline analysis.
2. Sleuth Kit for filesystem examination, inode analysis, metadata extraction, and timeline reconstruction.
3. Foremost for signature-based PDF recovery.
4. PhotoRec for independent PDF recovery and validation.

The investigation also incorporated Windows Registry analysis using reglookup and a NIST reference dataset.

The project established a documented evidence-preservation process using read-only originals, verified working copies, SHA-256 hashing, and evidence registration.

The results demonstrated that deleted information could be recovered using different forensic methods, that independent tools could corroborate examination results, and that filesystem and registry artifacts could be correlated into supported chronological findings.

A formal forensic report was produced with supporting screenshots, examination methodology, tool configurations, evidence records, findings, and technical limitations.

These implementations establish a substantial foundation for digital forensic investigation, evidence management, and professional forensic documentation.

Further implementation of two additional tools from the rubric's named list (FTK, EnCase, SIFT, or Volatility) would strengthen compliance with its specific enterprise tooling requirement.